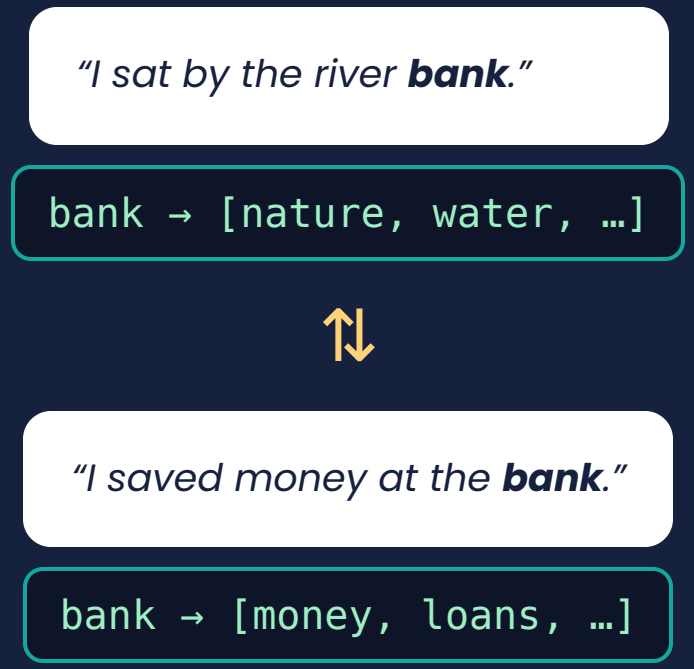
NATURAL LANGUAGE PROCESSING · LESSON 4

How AI Chatbots Understand Context

Transformers, attention, and using ready-made AI models.



WARM-UP

Same word, different meaning

Read these two sentences. What does “bank” mean in each one?

“*I sat by the river* ***bank****.*”

Here, “bank” means the land beside a river.

“*I saved money at the* ***bank****.*”

Here, “bank” means a place that keeps money.

You knew instantly from the **other words** in the sentence. In Lesson 3, our computer gave both “banks” the exact same numbers.

WHY THIS MATTERS

Why learn how chatbots work?



You already use them

AI chatbots help people write, study, translate, and code. Many students use them every week.



They are everywhere

Search engines, voice assistants, and translation apps all use the same technology you will learn today.



Know how they think

When you understand how chatbots work, you know when to trust them and when to double-check them.



Future careers

Building and testing these models is one of the fastest-growing jobs in technology.

THE BIG REVEAL

Chatbots are not magic

An AI chatbot writes by **predicting the next word**, again and again, very fast.

Step 1. It reads your message and everything written so far.

Step 2. It guesses the most likely next word and adds it.

Step 3. It repeats until the answer is complete.

To guess well, it must understand **context**. Today you will learn the trick that makes this possible: **attention**.

QUICK RECAP

Our NLP journey so far

Each lesson fixed a problem from the lesson before it.

L1
Clean text
Tokens, lemmas

→

L2
Count words
BoW, TF-IDF

→

L3
Word meaning
Embeddings

→

L4
Context
Transformers

Lesson 3's problem: one fixed list of numbers per word. Today's fix: numbers that **change with the sentence**.

LEARNING GOALS

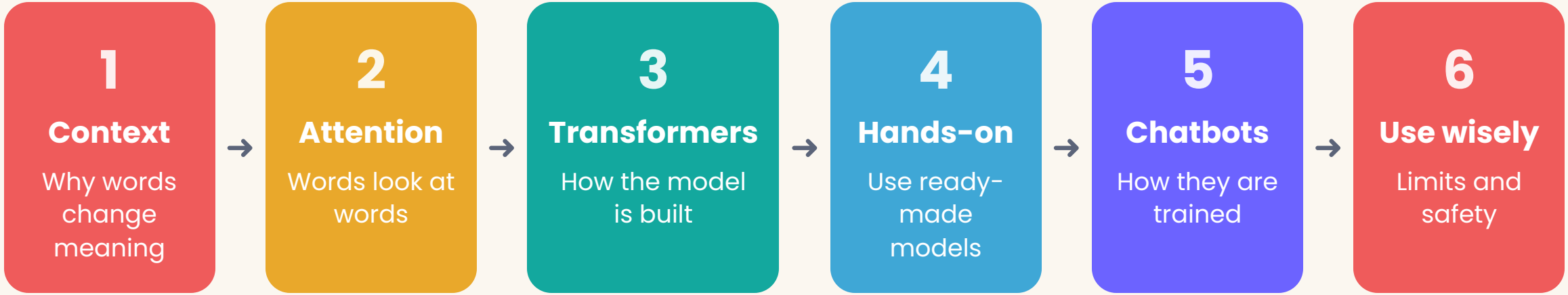
What we will learn today

1. Explain why the meaning of a word depends on its context.
2. Describe how attention helps words “look at” each other.
3. Explain how transformers split text into word pieces.
4. Tell the difference between BERT-style and GPT-style models.
5. Use ready-made AI models for real tasks in Python.
6. Use AI chatbots wisely by spotting hallucinations and bias.

ROADMAP

Today’s journey

Here is our plan for today’s lesson.



01

The Context Problem

Why one meaning per word is not enough.

CONTEXT

Context changes meaning

Words with many meanings

“bank” (river or money), “bat” (animal or sports), “light” (not heavy or a lamp).

Sentences with two readings

“I saw her duck.” Did she lower her head, or do you see her pet bird? We met this one in Lesson 1.

Words that point back

“The animal didn’t cross the street because **it** was too tired.” What does “it” mean?

In every case, you need the **other words** in the sentence to understand the meaning.

CONTEXT

From fixed numbers to flexible numbers

Lesson 3: word embeddings

The word “bank” always gets the **same** list of numbers, no matter the sentence.

```
river bank → bank = [same]
money bank → bank = [same]
```

Today: contextual embeddings

The word “bank” gets **different** numbers depending on the words around it.

```
river bank → bank = [nature]
money bank → bank = [money]
```

The question is: how does the computer know which words to look at? The answer is **attention**.

02

Attention

Every word looks at every other word.

ATTENTION

The big idea: attention

Attention lets each word look at all the other words in the sentence and decide which ones help explain its meaning.

Think of a class discussion

When a classmate asks about the science project, you pay more attention to the people talking about science and less to the others.

Words do the same thing

In “river bank,” the word “bank” pays close attention to “river.” That attention pulls its meaning toward nature and water.

ATTENTION

What does “it” mean?

ILLUSTRATION Darker purple means “it” pays more attention to that word.

The animal didn’t cross the street because it was too tired.

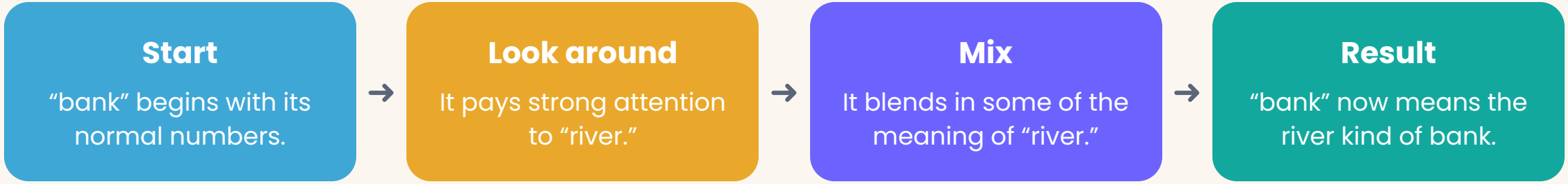
Here, “it” means the **animal**, because animals get tired.

The animal didn’t cross the street because it was too wide.

Change one word, and now “it” means the **street**, because streets are wide.

How attention changes a word's meaning

Attention builds a new, context-aware meaning for every word.



The result is called a **contextual embedding**: the numbers for a word that fit its sentence.

ATTENTION

Many heads and many layers

Many heads

Each layer has several **attention heads**. Like a team of detectives, each head looks for a different kind of clue: grammar, meaning, or who did what.

Many layers

Attention happens again and again in **layers**. Each layer builds a deeper understanding on top of the one before it.

The model we use today, DistilBERT, has **6 layers** with **12 attention heads** each.

03

Transformers

The design behind modern AI.

What is a transformer?

A **transformer** is a model built around attention. It was introduced in 2017 by Google researchers in a paper called “Attention Is All You Need.”



Almost every modern AI chatbot is built on the transformer design.

SIMULATED DEMO

Step 1: word pieces

SIMULATED DEMO

```
from transformers import AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained(MODELS["bert"])

print(tokenizer.tokenize("Tokenization is fun"))
print(tokenizer.tokenize("embeddings"))
```

EXAMPLE OUTPUT

```
['token', '##ization', 'is', 'fun']
['em', '##bed', '##ding', '##s']
```

Transformers split text into **word pieces**. A piece starting with **##** continues the word before it.

Common words stay whole. Rare or long words are split into smaller pieces the model already knows.

This fixes Lesson 3's problem with unknown words like "fr33." Any word can be built from pieces.

Two famous families: BERT and GPT

Feature	BERT family	GPT family
How it reads	Looks at words on **both sides** of each word.	Reads **left to right**, like writing a sentence.
Training game	Fill in a hidden word in the middle.	Guess the next word.
Best at	Understanding: sentiment, topics, search.	Writing: stories, answers, chatbots.
Today's model	DistilBERT (66 million parameters)	DistilGPT-2 (82 million parameters)

Parameters are the numbers a model learns during training. More parameters usually means a more powerful model.

How transformers learn: two games

BERT's game

Hide about 15% of the words, then guess them.

"*The cat sat on the* **?** ."

Possible answers: mat, floor, bed.

GPT's game

Read the words so far, then guess the next one.

"*The cat sat on the …*"

Possible answers: mat, floor, bed.

No human labels are needed. The text itself is the answer key, so models can learn from **billions** of sentences.

Pre-training and fine-tuning

1. Pre-training

Like your general education. The model reads huge amounts of text and learns how language works in general.

2. Fine-tuning

Like specialized training for a job. The model gets a little extra training on one task, like movie reviews.

Our sentiment model is DistilBERT, **fine-tuned** on thousands of movie review sentences labeled positive or negative.

04

Hands-On with Ready-Made Models

Real AI in just a few lines of code.

HANDS-ON

Meet Hugging Face and our models

Hugging Face

A free website and Python library that shares thousands of ready-made AI models. Its **pipeline()** function lets you use a model in about three lines of code.

Why DistilBERT?

Its creators report that it is about 40% smaller and 60% faster than BERT, while keeping about 97% of BERT's language understanding.

Task today	Model
Word pieces, fill in the blank, "bank" test	`distilbert-base-uncased`
Sentiment (positive or negative)	`distilbert-…-finetuned-sst-2-english`
Sorting text into any topics	`distilbert-base-uncased-mnli`
Predicting the next word	`distilgpt2`

SIMULATED DEMO

Demo 1: fill in the blank

SIMULATED DEMO

```
from transformers import pipeline

fill_mask = pipeline("fill-mask", model=MODELS["bert"])

results = fill_mask("The capital of France is [MASK].")
print(results[0]["token_str"])
```

EXAMPLE OUTPUT

```
paris
```

[MASK] marks the hidden word, just like BERT's training game.

The model fills the blank using the words on **both sides**.

In the notebook, you will see its top five guesses with a confidence score for each.

SIMULATED DEMO

Demo 2: sentiment analysis

SIMULATED DEMO

```
# MODELS holds the model names (see the notebook)
sentiment = pipeline("text-classification",
                     model=MODELS["sentiment"])

sentiment([
    "I love this movie! The acting was wonderful.",
    "The food was cold and the service was slow.",
])
```

EXAMPLE OUTPUT

```
POSITIVE  (very confident)
NEGATIVE  (very confident)
```

This model never saw these sentences before, but it understands the feeling behind them.

Challenge: In the notebook, test “The movie wasn’t boring at all.” Bag of Words would see the word “boring” and get confused.

Does the transformer handle the word “wasn’t”? Run it and find out!

SIMULATED DEMO

Demo 3: sorting into any topics

SIMULATED DEMO

```
zero_shot = pipeline("zero-shot-classification",
                     model=MODELS["zero_shot"])

text = "The basketball team won the championship game."
topics = ["sports", "food", "homework", "weather"]

result = zero_shot(text, candidate_labels=topics)
print("Best topic:", result["labels"][0])
```

EXAMPLE OUTPUT

```
Best topic: sports
```

Zero-shot means the model sorts text into topics it was **never trained on**.

You choose the topics. You can change them anytime, without any new training.

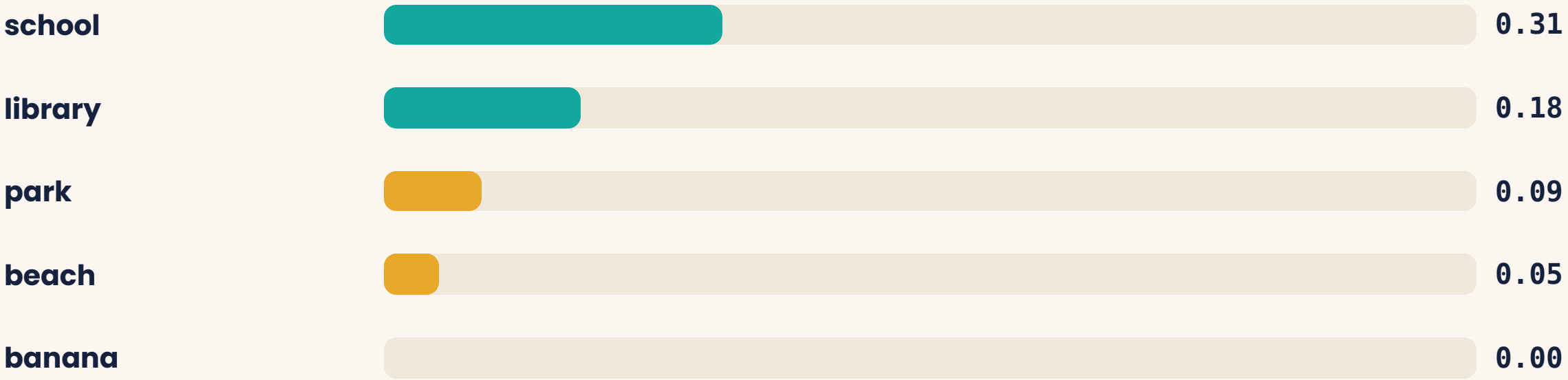
In the notebook, you will sort real school announcements, like class suspensions and canteen menus.

DEMO 4

How chatbots choose the next word

ILLUSTRATION A GPT model gives every possible next word a probability. These numbers are made up to show the idea.

“The students went to the …”



In the notebook, you will see the real top guesses from DistilGPT-2 for this exact sentence.

SIMULATED DEMO

Demo 5: solving the “bank” problem

SIMULATED DEMO

```
sentences = [
    "I sat on the bank of the river and ...",
    "We had a picnic on the grassy bank ...",
    "I deposited my money at the bank ...",
    "The bank gave her a loan to ...",
]

# The contextual numbers for each bank
vectors = [word_vector(s, "bank") for s in sentences]
```

WHAT TO LOOK FOR

```
Each 'bank' gets its own 768 numbers.
Do river banks match river banks?
Do money banks match money banks?
```

In Lesson 3, all four “banks” had the **same** numbers, so every similarity score was exactly 1.00.

Now each “bank” gets numbers that fit its sentence.

In the notebook, you will print a similarity table and check whether the river and money meanings separate.

EXPERIMENT

Transformer vs. Bag of Words for spam

In Lesson 2, Bag of Words caught spam with **98.6%** accuracy. Now we test a transformer on the exact same messages.

Make a prediction

Will the transformer beat 98.6%? Will it miss fewer spam messages? Write your guess before running the notebook.

Then think about cost

Bag of Words runs in a blink. The transformer takes minutes to read 5,572 messages. Is the difference worth it for a phone?

Real engineers always ask two questions: “Is it more accurate?” and “Is it worth the extra cost?”

05

How Chatbots Are Made

From a word guesser to a helpful assistant.

The three steps to build a chatbot

1

Pre-training

Read a huge amount of text and learn to predict the next word.

→

2

Instruction tuning

Practice on examples of questions with good answers.

→

3

Human feedback

People rate answers, and the model learns to give more helpful and safer ones.

Step 1 makes a model that can continue text. Steps 2 and 3 turn it into an assistant that answers your questions.

CHATBOTS

Small models vs. giant models

Our classroom model

82 million

parameters in DistilGPT-2

GPT-3 (2020)

175 billion

parameters, over 2,000 times more

Modern chatbots use the **same basic ideas** as our small models. The big difference is size, data, and training.

06

Using AI Wisely

Powerful tools need careful users.

LIMITS

AI can make things up

A **hallucination** is when an AI writes something that sounds confident and correct, but is actually false.

Why it happens

A language model predicts **likely** words. It does not look up facts or check if its answer is true.

What you should do

Always check important facts, like dates, names, and numbers, with reliable sources such as textbooks or official websites.

In the notebook, you will ask our small model simple questions and fact-check its answers.

LIMITS

Other things to watch out for



Bias

Models learn from human writing, so they can repeat stereotypes, just like the embeddings in Lesson 3.



Privacy

Never type passwords, addresses, or other private information into a chatbot.



Energy

Training giant models uses a lot of electricity and computing power.



Honesty

Use AI to help you learn and understand, not to copy. Follow your school's rules on AI use.

USE AI WISELY

The smart AI user checklist

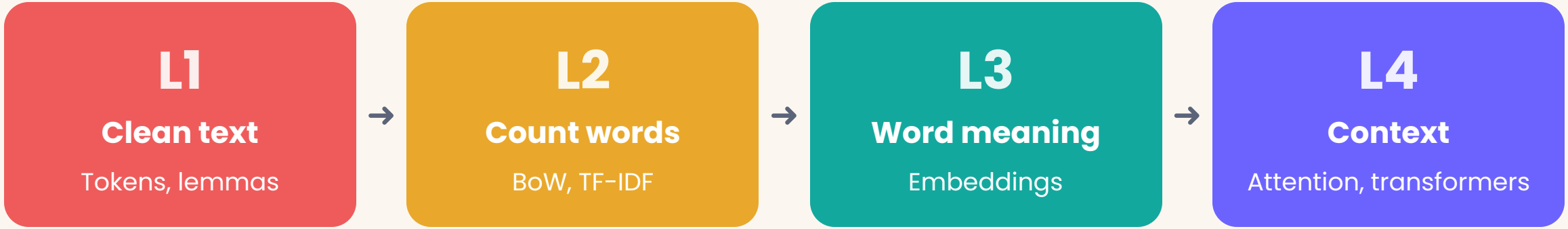
Before you trust an AI answer, go through this checklist.

- ✓ **Check the facts.** Verify important information with a reliable source.
- ✓ **Protect your privacy.** Keep personal details out of your prompts.
- ✓ **Think for yourself.** Use AI answers as a starting point, not the final word.
- ✓ **Be honest.** Say when and how you used AI in your work.

YOUR JOURNEY

Look how far you have come

You now know the main building blocks behind modern AI chatbots.



Every chatbot you use splits your words into tokens, turns them into numbers, and uses attention to understand context.

SUMMARY

Key takeaways

1. The meaning of a word depends on its context.
2. Attention lets each word look at the other words to understand its meaning.
3. Transformers split text into word pieces, so they can handle any word.
4. BERT-style models understand text; GPT-style models write text.
5. Models are pre-trained on huge text, then fine-tuned for specific tasks.
6. Chatbots write by predicting the next word, one word at a time.
7. AI can hallucinate and be biased, so always check important facts.

CHECK YOUR UNDERSTANDING

Quick quiz

Q1

What problem do contextual embeddings fix?

A. Words with spelling mistakes
B. One word having one fixed meaning in every sentence
C. Texts that are too long

Q2

What does attention let a word do?

A. Look at other words to understand its meaning
B. Delete stop words
C. Count how often it appears

Q3

Which training game does GPT play?

A. Fill in a hidden word in the middle
B. Guess the next word
C. Sort words A to Z

Q4

What is fine-tuning?

A. Training a model from zero
B. Extra training on a specific task
C. Making the model smaller

Q5

An AI says a false fact with confidence. What is this called?

A. A hallucination
B. A token
C. Attention

CHECK YOUR UNDERSTANDING

Quiz answers

Q1

Answer: B

The same word now gets different numbers in different sentences.

Q2

Answer: A

It decides which other words matter most for its meaning.

Q3

Answer: B

GPT reads left to right and predicts what comes next.

Q4

Answer: B

It adds a little extra training for one job, like movie reviews.

Q5

Answer: A

The model predicts likely words without checking facts.

ACTIVITY

Try it yourself

Activity 1: Human attention

Write a sentence with “it” on the board. Students draw arrows from “it” to the words that explain it. Then change one word and draw again.

Activity 2: Beat the blank

Teams write fill-in-the-blank sentences. Everyone guesses first, then the model guesses. Who wins more rounds: the class or the AI?

Activity 3: Hallucination hunt

Ask the model five factual questions. Fact-check each answer with a textbook and record how many were wrong.

Bonus: How could a chatbot be trained to understand Filipino and Taglish better?

NATURAL LANGUAGE PROCESSING

Thank you for listening!

You now understand the ideas behind modern AI chatbots.
What questions do you have?

`attention` `is` `all` `you` `need` `!`